Exponents & Polynomials

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 6$ in evaluating functions, evaluate $f(5)$.

Final Answer: 36. Substitute $x = 5$: $f(5) = 6(5) + 6 = 36$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 2x + 7$ in review: graphing linear relationships from a table of values, evaluate $f(2)$.

Final Answer: 11. Substitute $x = 2$: $f(2) = 2(2) + 7 = 11$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for review: graphing linear relationships from a table of values consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 3x + 8$ in characteristics from a graph, evaluate $f(2)$.

Final Answer: 14. Substitute $x = 2$: $f(2) = 3(2) + 8 = 14$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for characteristics from a graph consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 24.5. The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 4x + 9$ in inverse relations and functions, evaluate $f(2)$.

Final Answer: 17. Substitute $x = 2$: $f(2) = 4(2) + 9 = 17$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for inverse relations and functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 16.5. The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Radicals & Exponents · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

[Linear Equations & Lines · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Systems & Inequalities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A parabola has its vertex at $(3, -4)$ and passes through the point $(1, 0)$. (a) Write the equation of the parabola in standard form $y = ax^2 + bx + c$. (b) Determine the coordinates of the other x-intercept.

(a) Use vertex form: With vertex $(h, k) = (3, -4)$: $y = a(x - 3)^2 - 4$ Substitute $(x, y) = (1, 0)$ to find a: $0 = a(1 - 3)^2 - 4 \Rightarrow 0 = 4a - 4 \Rightarrow 4a = 4 \Rightarrow a = 1$. Expand to standard form: $y = 1(x^2 - 6x + 9) - 4 = \mathbf{x^2 - 6x + 5}$. (b) Find the other x-intercept: By symmetry, the x-intercepts are equidistant from the axis of symmetry $x = 3$. Since one intercept is at $x = 1$ (distance $3 - 1 = 2$), the other intercept is at $x = 3 + 2 = 5$, giving **(5, 0)**. (V.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Simplify the expression $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4}$ and write your answer using only positive exponents.

Step 1: Invert the fraction to remove the negative exponent: $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4} = \left(\frac{81z^{12}}{16x^8y^{-4}}\right)^{3/4} = \left(\frac{81y^4z^{12}}{16x^8}\right)^{3/4}$. Step 2: Take the 4th root of each factor: $\left(\frac{81y^4z^{12}}{16x^8}\right)^{1/4} = \frac{81^{1/4}(y^4)^{1/4}(z^{12})^{1/4}}{16^{1/4}(x^8)^{1/4}} = \frac{3yz^3}{2x^2}$. Step 3: Cube the result.

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14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In a course grading system, Homework is weighted 20%, Midterm Exam is weighted 35%, and Final Exam is weighted 45%. A student scored 92 on Homework and 76 on the Midterm Exam. What is the minimum score the student must achieve on the Final Exam to earn an overall course average of at least 84?

Step 1: Set up the weighted average inequality with final exam score F:
$0.20(92) + 0.35(76) + 0.45(F) \geq 84$. Step 2: Compute the accumulated points:
$0.20(92) = 18.4, \quad 0.35(76) = 26.6 \Rightarrow 18.4 + 26.6 = 45.0$. Step 3: Isolate F:
$45.0 + 0.45F \geq 84 \Rightarrow 0.45F \geq 84 - 45 = 39 \Rightarrow F \geq \frac{39}{0.45} = \frac{3900}{45} = \frac{260}{3} \approx 86.67$. The student must score at least **86.7** (or $\mathbf{\frac{260}{3}}$) on the Final Exam.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 8.01 Factoring Polynomials**

Solve the radical equation $\sqrt{2x+7} - x = 2$ for all real solutions and confirm extraneous roots.

Isolate the radical term: $\sqrt{2x+7} = x + 2$. Square both sides: $2x + 7 = (x+2)^2 = x^2 + 4x + 4$.
$x^2 + 2x - 3 = 0 \Rightarrow (x+3)(x-1) = 0 \Rightarrow x = -3 \quad \text{or} \quad x = 1$. Test in the original un-squared equation:Check $x = -3$:
$\sqrt{2(-3)+7} - (-3) = \sqrt{1} + 3 = 1 + 3 = 4 eq 2$. (Extraneous ✗) Check $x = 1$:
$\sqrt{2(1)+7} - 1 = \sqrt{9} - 1 = 3 - 1 = 2$. (Valid ✓) The only valid real solution is $x = 1$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 9.01 Radicals & Exponents**

If $x - 2$ is a factor of the polynomial $P(x) = x^3 - kx^2 + 3x - 10$, find the value of k, and determine all real roots of $P(x)$.

Step 1: By the Factor Theorem, $x - 2$ is a factor $\Leftrightarrow P(2) = 0$:
$P(2) = (2)^3 - k(2)^2 + 3(2) - 10 = 0 \Rightarrow 8 - 4k + 6 - 10 = 0 \Rightarrow 4 - 4k = 0 \Rightarrow \mathbf{k = 1}$. Step 2: Substitute $k = 1$ into $P(x) = x^3 - x^2 + 3x - 10$ and divide by $(x - 2)$ using polynomial long division or synthetic division:
$x^3 - x^2 + 3x - 10 = (x-2)(x^2 + x + 5)$. Step 3: Solve $x^2 + x + 5 = 0$. Its discriminant is \(\Delta = (1)^2 - 4(1)(5) = 1 - 20 = -19

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 1.01 Linear Equations & Lines**

Solve the quadratic equation $3x^2 + 10x - 8 = 0$ by factoring.

Step 1: Find two integers whose product is $a \cdot c = 3(-8) = -24$ and whose sum is $b = 10$. The factors are $+12$ and -2. Step 2: Split the middle term and factor by grouping:
$3x^2 + 12x - 2x - 8 = 0 \Rightarrow 3x(x+4) - 2(x+4) = 0 \Rightarrow (3x-2)(x+4) = 0$ Step 3: Set each factor to zero:
$3x - 2 = 0 \Rightarrow x = \frac{2}{3}, \quad x + 4 = 0 \Rightarrow x = -4$ The solutions are $\mathbf{x = \frac{2}{3}}$ and $\mathbf{x = -4}$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Rational Expressions

A bivariate regression model for student preparation hours x and exam grade y yields the line of best fit $\hat{y} = 4.8x + 52$. A student who studied for 8 hours scored 94 on the exam. (a) What is the model's predicted score for this student? (b) Compute and interpret the student's regression residual.

(a) Predicted Score $\hat{y}$: Substitute $x = 8$ into the regression equation: $\hat{y} = 4.8(8) + 52 = 38.4 + 52 = \mathbf{90.4}$. (b) Residual Calculation & Interpretation: $\text{Residual} = y - \hat{y} = 94 - 90.4 = \mathbf{+3.6}$. Because the residual is positive (+3.6), the student's actual score was 3.6 points higher than the score predicted by the regression line (the model underestimated the student's performance).

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Statistics & Data Analysis

Level 3 · AdvancedWhat is the set of all real solutions to the radical equation $\sqrt{2x - 5} = x - 4$?

Square both sides: $2x - 5 = (x - 4)^2 = x^2 - 8x + 16 \Rightarrow x^2 - 10x + 21 = 0 \Rightarrow (x - 3)(x - 7) = 0.$ Check prospective roots in original equation:For $x = 3$: $\sqrt{2(3) - 5} = \sqrt{1} = 1$, but right side is $3 - 4 = -1$. Since $1 eq -1$, $x = 3$ is extraneous.For $x = 7$: $\sqrt{2(7) - 5} = \sqrt{9} = 3$, and right side is $7 - 4 = 3$. Valid!The only real solution is $x = 7$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Quadratic Functions

Convert the quadratic function $y = 2x^2 - 12x + 11$ into vertex form $y = a(x - h)^2 + k$. State the coordinates of the vertex and the line of symmetry.

Step 1: Factor the leading coefficient $a = 2$ from the x-terms: $y = 2(x^2 - 6x) + 11$ Step 2: Complete the square inside parentheses. Half of -6 is -3, and $(-3)^2 = 9$: $y = 2(x^2 - 6x + 9 - 9) + 11 = 2(x - 3)^2 - 2(9) + 11 = 2(x - 3)^2 - 18 + 11$ Step 3: Simplify to vertex form: $y = \mathbf{2(x - 3)^2 - 7}$ The vertex is $\mathbf{(3, -7)}$ and the axis of symmetry is $\mathbf{x = 3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 4$ in functions and relations, evaluate $f(2)$.

a. 12 ✓

b. 16

c. 8

d. 16

Substitute $x = 2$: $f(2) = 4(2) + 4 = 12$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 5$ in domain and range and set notation, evaluate $f(2)$.

a. 15 ✓

b. 20

c. 10

d. 19

Substitute $x = 2$: $f(2) = 5(2) + 5 = 15$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for functions and relations consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for domain and range and set notation consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the sum of all real solutions to the equation $(x^2 - 5x + 5)^{x^2 - 14x + 48} = 1$?

a. 19

b. **21** ✓

c. 22

d. 24

An equation of the form $[P(x)]^{Q(x)} = 1$ for real numbers x holds if and only if at least one of three mutually exclusive cases is satisfied: **Case 1: The exponent is zero and the base is nonzero.** $Q(x) = x^2 - 14x + 48 = 0 \Rightarrow (x-6)(x-8) = 0 \Rightarrow x = 6$ or $x = 8$. We must verify that the base $P(x) = x^2 - 5x + 5 eq 0$: - For $x = 6$: $P(6) = 6^2 - 5(6) + 5 = 11 eq 0$. Valid. - For $x = 8$: $P(8) = 8^2 - 5(8) + 5 = 29 eq 0$. Valid. **Case 2: The base is equal to 1.**.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the polynomial $P(x) = ax^3 + bx^2 + cx + d$. If $P(x)$ has roots at $x = 1$, $x = 2$, and $x = 3$, and $P(0) = 12$, find the value of the expression $\frac{a+b+c}{d}$.

a. $-1/2$ ✓

b. -2

c. -1

d. 2

Since the roots of the polynomial $P(x) = ax^3 + bx^2 + cx + d$ are $x = 1$, $x = 2$, and $x = 3$, we can write $P(x)$ in factored form as $P(x) = a(x-1)(x-2)(x-3)$. We are given that $P(0) = 12$. Substituting $x = 0$ into the factored form: $P(0) = a(0-1)(0-2)(0-3) = a(-1)(-2)(-3) = -6a$. Since $P(0) = 12$, we have $-6a = 12$, which implies $a = -2$. Now we expand the factored form to find the coefficients b, c, and d in terms of a.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $2^{a+b} = 20$ and $2^{a-b} = 5$, what is the value of $4^a - 4^b$?

a. 15

b. 96 ✓

c. 104

d. 375

We are given the two exponential equations: 1) $2^{a+b} = 20$ 2) $2^{a-b} = 5$ Recall the exponent product and quotient rules: $2^x \cdot 2^y = 2^{x+y}$ and $\frac{2^x}{2^y} = 2^{x-y}$. Multiplying the two equations gives: $2^{a+b} \cdot 2^{a-b} = 2^{(a+b)+(a-b)} = 2^{2a}$ $2^{2a} = 20 \times 5 = 100$ Since $4^a = (2^2)^a = 2^{2a}$, we have $4^a = 100$. Dividing the two equations gives: $\frac{2^{a+b}}{2^{a-b}} = 2^{(a+b)-(a-b)} = 2^{2b}$ $2^{2b} = \frac{20}{5} = 4$ Since $4^b = (2^2)^b = 2^{2b}$, we have.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Using the Remainder Theorem, find the remainder when $f(x) = x^3 + 5x^2 + 0x + 4$ is divided by $(x - 3)$.

a. 78

b. 77

c. **76** ✓

d. 74

By the Remainder Theorem, the remainder equals $f(3) = (3)^3 + 5(3)^2 + 0(3) + 4 = 76$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The polynomial $P(x) = x^3 + ax^2 - 5x + b$ has roots $x = 1$ and $x = -2$. If $P(x)$ is divided by $(x - c)$, the remainder is 0. What is the value of c?

a. -3

b. -1

c. 2

d. 3 ✓

Since $x = 1$ and $x = -2$ are roots of $P(x)$, by the Factor Theorem, $(x - 1)$ and $(x + 2)$ are factors of $P(x)$. Therefore, $P(x)$ can be written as $(x - 1)(x + 2)(x - r)$ for some third root r. Expanding $(x - 1)(x + 2)$ gives $x^2 + x - 2$. So, $P(x) = (x^2 + x - 2)(x - r) = x^3 - rx^2 + x^2 - rx - 2x + 2r = x^3 + (1 - r)x^2 + (-r - 2)x + 2r$. Comparing the coefficients with $P(x) = x^3 + ax^2 - 5x + b$: 1. The coefficient of x^2: $a = 1 - r$. 2. The coefficient of x: $-5 = -r - 2$. From this equation, we can solve for r.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given the expression $\sqrt[3]{64}$, what is the principal cube root of 64?

a. **4** ✓

b. 8

c. 16

d. 2

The cube root of a number 'x', denoted as $\sqrt[3]{x}$, is a number 'y' such that $y^3 = x$. We are looking for the number that, when multiplied by itself three times, equals 64. Let's test the options: - $2^3 = 2 \times 2 \times 2 = 8$ (Incorrect) - $4^3 = 4 \times 4 \times 4 = 16 \times 4 = 64$ (Correct) - $8^3 = 8 \times 8 \times 8 = 64 \times 8 = 512$ (Incorrect) - $16^3 = 16 \times 16 \times 16 = 256 \times 16 = 4096$ (Incorrect) Therefore, the principal cube root of 64 is 4.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which expression is equivalent to $(16x^8)^{\frac{1}{4}}$ for all $x > 0$?

a. $2x^2$ ✓

b. $4x^2$

c. $2x^4$

d. $4x^4$

Apply the power of a product rule, $(ab)^n = a^n b^n$: $(16x^8)^{\frac{1}{4}} = 16^{\frac{1}{4}} \cdot (x^8)^{\frac{1}{4}}$ 1. Evaluate the numerical base: $16^{\frac{1}{4}} = \sqrt[4]{16} = 2$, because $2^4 = 16$. A common error is dividing 16 by 4 to get 4. 2. Apply the power of a power rule, $(x^a)^b = x^{ab}$: $(x^8)^{\frac{1}{4}} = x^{8 \cdot \frac{1}{4}} = x^2$ Combining the terms yields $2x^2$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Evaluate the expression: $\sqrt{64} - \sqrt[3]{27}$

a. **5** ✓

b. 6

c. 7

d. 8

The square root of 64 is 8, since $8^2 = 64$. The cube root of 27 is 3, since $3^3 = 27$. Therefore, the expression simplifies to $8 - 3 = 5$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A sequence is defined by the recurrence relation $a_{n+1} = \frac{a_n + k}{a_n}$, where k is a positive constant and $a_1 = 1$. If the sequence converges to a limit L, and $a_3 = \frac{5}{3}$, find the value of k.

a. 1

b. **2** ✓

c. 3

d. 4

The recurrence relation is $a_{n+1} = \frac{a_n + k}{a_n}$. We are given $a_1 = 1$. We can find the first few terms of the sequence in terms of k: $a_2 = \frac{a_1 + k}{a_1} = \frac{1 + k}{1} = 1 + k$ $a_3 = \frac{a_2 + k}{a_2} = \frac{(1 + k) + k}{1 + k} = \frac{1 + 2k}{1 + k}$ We are given that $a_3 = \frac{5}{3}$. Therefore, we can set up the equation: $\frac{1 + 2k}{1 + k} = \frac{5}{3}$ Cross-multiply to solve for k: $3(1 + 2k) = 5(1 + k)$ $3 + 6k = 5 + 5k$ $6k - 5k = 5 - 3$ $k = 2$ Alternatively, if the s.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A geometric sequence has its first term a and common ratio r. The sum of the first three terms is 14, and the sum of the first six terms is 126. If a and r are positive constants, find the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

Let the geometric sequence be denoted by a_n, where $a_1 = a$ and the common ratio is r. The sum of the first n terms of a geometric sequence is given by $S_n = \frac{a(r^n - 1)}{r - 1}$ (for $req1$). We are given: 1. The sum of the first three terms is 14: $S_3 = \frac{a(r^3 - 1)}{r - 1} = 14$ 2. The sum of the first six terms is 126: $S_6 = \frac{a(r^6 - 1)}{r - 1} = 126$ We can rewrite S_6 using the difference of squares factorization for $r^6 - 1 = (r^3)^2 - 1^2 = (r^3 - 1)(r^3 + 1)$. So.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The second and fifth terms of an arithmetic sequence are $x - 3$ and $2x + 6$, respectively, where x is a constant. If the common difference of the sequence is $x - 5$, what is the sum of the first 10 terms of the sequence?

a. 65

b. 335 ✓

c. 405

d. 475

In an arithmetic sequence with first term a_1 and common difference d, the n-th term is given by $a_n = a_1 + (n - 1)d$. The difference between the fifth term a_5 and the second term a_2 is: $a_5 - a_2 = (a_1 + 4d) - (a_1 + d) = 3d$ Substitute the given expressions $a_5 = 2x + 6$, $a_2 = x - 3$, and $d = x - 5$: $(2x + 6) - (x - 3) = 3(x - 5)$ Simplify both sides: $x + 9 = 3x - 15$ $24 = 2x \Rightarrow x = 12$ Now, evaluate the parameters of the sequence: - Common difference: $d = 12 - 5 = 7$ - Second ter.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An arithmetic sequence has its first term a_1 and common difference d. The sum of the first n terms is denoted by S_n. If $S_{10} = 155$ and $S_{20} = 610$, find the value of the common difference d.

a. 3

b. **4** ✓

c. 5

d. 6

The sum of the first n terms of an arithmetic sequence is given by $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$. We are given $S_{10} = 155$. Substituting $n = 10$ into the formula: $155 = \frac{10}{2}(2a_1 + (10 - 1)d)$ $155 = 5(2a_1 + 9d)$ Dividing by 5: $31 = 2a_1 + 9d$ (Equation 1) We are also given $S_{20} = 610$. Substituting $n = 20$ into the formula: $610 = \frac{20}{2}(2a_1 + (20 - 1)d)$ $610 = 10(2a_1 + 19d)$ Dividing by 10: $61 = 2a_1 + 19d$ (Equation 2) Now we have a system of two linear equations with two variables.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A sequence is defined by the recursive formula $a_n = 3a_{n-1}$ with the first term $a_1 = 2$. What is the value of the third term, a_3?

a. 6

b. **18** ✓

c. 2

d. 12

The first term is given as $a_1 = 2$. The recursive formula is $a_n = 3a_{n-1}$. To find the second term, a_2, we use $n = 2$: $a_2 = 3a_{2-1} = 3a_1 = 3 \times 2 = 6$. To find the third term, a_3, we use $n = 3$: $a_3 = 3a_{3-1} = 3a_2 = 3 \times 6 = 18$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Rational Expressions

Which formula gives the nth term of an arithmetic sequence?

a. $a_n = a_1 \cdot r^{n-1}$

b. $a_n = a_1 \cdot n$

c. $a_n = a_1 + nd$

d. $a_n = a_1 + (n-1)d$ ✓

Applying Sequences & Series principles, Arithmetic: $a_n = a_1 + (n-1)d$. Therefore, the verified correct option is $a_n = a_1 + (n-1)d$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Statistics & Data Analysis

What is the common ratio of $81, 27, 9, 3, \ldots$?

a. $\frac{1}{3}$ ✓

b. $\frac{1}{9}$

c. 3

d. -3

Applying Sequences & Series principles, $\frac{27}{81} = \frac{1}{3}$. Therefore, the verified correct option is $\frac{1}{3}$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Quadratic Functions

Find the 6th term of a geometric sequence with $a_1 = 2$ and $r = 3$.

a. 729

b. 162

c. 243

d. 486 ✓

Applying Sequences & Series principles, $a_6 = 2 \cdot 3^5 = 2 \cdot 243 = 486$. Therefore, the verified correct option is 486.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Functions & Transformations

If $-4 \le x \le 6$, what is the range of $3x - 2$?

a. $[-15, 16]$

b. $[-14, 16]$ ✓

c. $[-13, 15]$

d. $[-14, 17]$

Since $3 > 0$, the expression is increasing: at $x = -4$ it's -14; at $x = 6$ it's 16. Range: $[-14, 16]$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the sum of the arithmetic series with first term 6, common difference 2, and 8 terms.

a. 96

b. 102

c. 112

d. **104** ✓

$a_8 = 6 + (8 - 1)(2) = 20$. $S_n = \frac{n(a_1 + a_n)}{2} = \frac{8(6 + 20)}{2} = 104$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Systems & Inequalities

Sum the arithmetic series with first term 2, last term 50, and 25 terms.

a. 650 ✓

b. 600

c. 625

d. 675

$S_{25} = \frac{25}{2}(2 + 50) = \frac{25 \cdot 52}{2} = 650$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Sequences & Series

Solve the system $3x + 1y = -1$ and $3x - 4y = 4$. What is x?

a. 2

b. -1

c. 1

d. **0** ✓

Solving simultaneously gives $x = 0$, $y = -1$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Factoring Polynomials

A geometric sequence has first term 3 and common ratio 3. Find the 4th term.

a. 27

b. 84

c. -81

d. **81** ✓

$a_4 = a_1 r^{n-1} = 3 \cdot (3)^{4-1} = 81$.

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